



COMP3153/9153

Algorithmic Verification

Lecture 16: Expressiveness of Timed Logics

Unbounded time

Situation so far

Interpreting logical formulas in Kripke structures

- Finite states
- Discrete time

Unbounded time

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- Discrete events

Unbounded (discrete) time model: $\mathbb{N}$

$$f : \mathbb{N} \rightarrow 2^{\text{MP}} \quad (\text{flow or signal})$$

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Unbounded (discrete) time model: $\mathbb{N}$

$$f : \mathbb{N} \rightarrow 2^{\mathbf{MP}} \quad (\text{flow or signal})$$

- Negative (discrete) time: $\mathbb{Z}$
- Dense time: $\mathbb{R}_{\geq 0}$
- Negative and dense time: $\mathbb{R}$

Classic temporal predicate logic

First-order logic (FO(<>):

$$\varphi ::= x < y \mid P(x) \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \vee \varphi_2 \mid \neg \varphi \mid \forall x \varphi \mid \exists x \varphi$$

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Monadic second-order logic (MSO(<>):

$$\varphi ::= x < y \mid P(x) \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \vee \varphi_2 \mid \neg \varphi \mid \forall x \varphi \mid \exists x \varphi \mid \forall P \varphi \mid \exists P \varphi$$

Relationship with other temporal logics

LTl is subsumed by FO($<$), for example

$$P \text{ U } Q \quad \equiv \quad \exists x (Q(x) \wedge \forall y . (y < x) \rightarrow P(y)).$$

Büchi's theorem

Flows over non-negative integer time are infinite words over the alphabet 2^{MP} .

Theorem (Büchi 1960)

Any MSO($<$) formula φ can be effectively translated into an equivalent automaton A_φ .

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Any $\text{MSO}(<)$ formula φ can be effectively translated into an equivalent automaton A_φ .

Corollary (Church 1960)

The satisfiability problems for $\text{MSO}(<)$ and $\text{FO}(<)$ are decidable.

What about complexity?

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Any MSO($<$) formula φ can be effectively translated into an equivalent automaton A_φ .

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The satisfiability problems for MSO($<$) and FO($<$) are decidable.

What about complexity?

Theorem (Stockmeyer 1974)

The satisfiability problem for FO($<$) has non-elementary complexity.

Temporal logics: LTL

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The satisfiability problem for LTL is PSPACE-complete.

Theorem (Kamp 1968)

LTL is as expressive as $FO(<)$ (over $\mathbb{N}$).



Temporal logics: LTL+Past

Requirements such as

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Linear Temporal Logic with past operators (LTL+Past):

θ	$::=$	P	$ $	$\theta_1 \wedge \theta_2$	$ $	$\theta_1 \vee \theta_2$	$ $	$\neg\theta$	
			$ $	F θ					(θ occurs in the Future)
			$ $	G θ					(θ occurs always (Globally))
			$ $	θ_1 U θ_2					(θ_1 holds Until θ_2)
			$ $	P θ					(θ occurred in the Past)
			$ $	H θ					(θ has always occurred (Historically))
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G (ACK $\rightarrow$ **P** REQ)

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LTL+Past is the original tense logic developed by Prior.

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Is it more expressive?

Kamp's theorem says no.

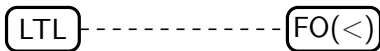


Other temporal logics

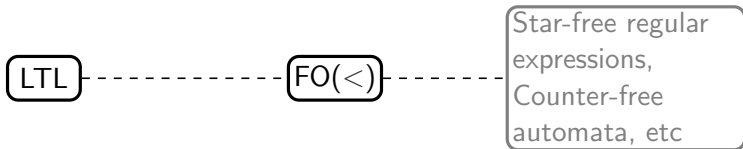
Towards MSO(<):

- ETL: LTL with existential quantification
- μ TL: LTL with fix-points
- LDL: LTL with regular expressions
- ...

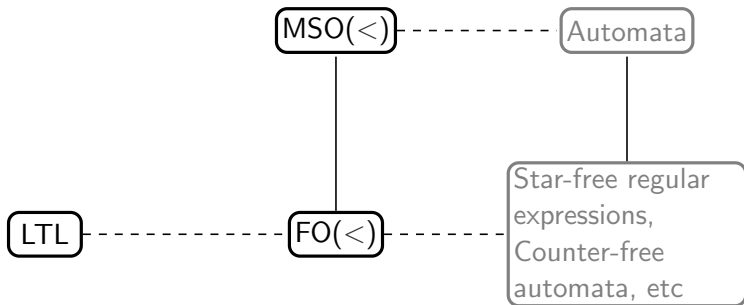
Expressiveness in the classical case



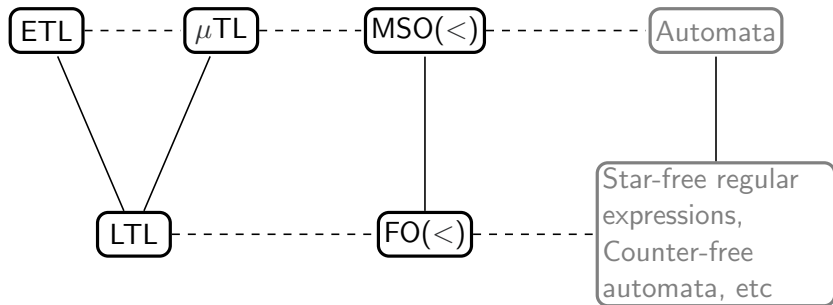
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Qualitative extensions: Integer time

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Classical:

- Flows: $f : \mathbb{N} \rightarrow 2^{\text{MP}}$
- Predicate logics: $\text{FO}(<)$ and $\text{MSO}(<)$
- Temporal logics: LTL, LTL+Past, ...

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Classical+Past:

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Decidability and Expressiveness results (with LTL+Past) still go through.

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How to handle semantics?

Pointwise (time-stamped events):

$$(a_0, \tau_0), (a_1, \tau_1), (a_2, \tau_2), \dots$$

where $\tau_0 < \tau_1 < \dots \in \mathbb{Q}$ (or $\mathbb{R}$)
and $a_i \subseteq \mathbf{MP}$

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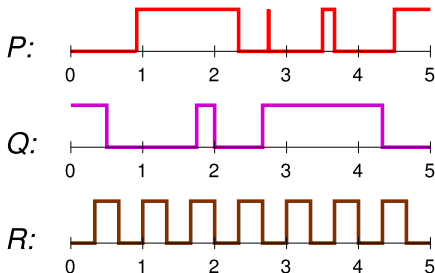
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Some bad news:

Theorem (Shelah 1975)

$MSO(<)$ is undecidable over $[0, 1)$.



Kamp's theorem again

Some great news:

Kamp's theorem applies over all domains (rationals need Stavi connectives)

Theorem (Kamp 1968; Gabbay et al. 1980)

LTL+Past is as expressive as $FO(<)$ (over $\mathbb{R}$).

Separation of temporal logics

A temporal logic formula is

- **pure past** if it is invariant on flows that agree on the past
- **pure present** if is invariant on flows that agree on the present
- **pure future** if is invariant on flows that agree on the future

A temporal logic is **separable** if all its formulas are equivalent to a boolean combination of pure past, present and future formulas.

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Lemma

LTL+Past is separable.

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Theorem (Gabbay 1981)

A temporal logic is expressively complete if and only if it is separable.



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Proof sketch:

⇒: Model-theoretic argument.



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Proof sketch:

$\Rightarrow$: Model-theoretic argument.

$\Leftarrow$: Use separation to eliminate quantified variables.

Qualitative extensions: Summary

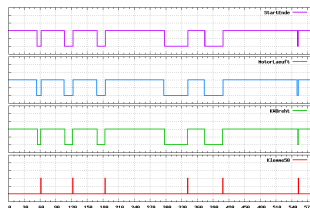
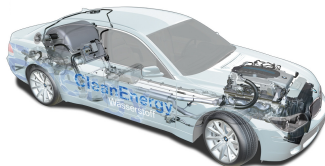
	FO(<) decidable?	MSO(<) decidable?	LTL expressively complete?
Classical	Yes	Yes	Yes
Integer time	Yes	Yes	Yes
Rational time	Yes	Yes	Sort of
Real time	Yes	No	Yes

Motivation

Timed systems are everywhere:

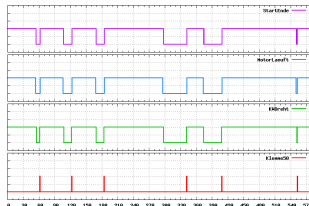
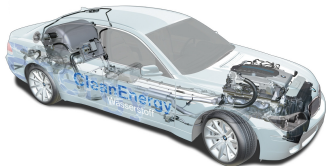
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Want to specify:

*“If I press the brake pedal then
the pads will be applied within 0.1ms.”*

Motivation

Timed systems also occur in:

- Hardware circuits
- Communication protocols
- Cell phones
- Plant controllers
- Aircraft navigation systems
- ...

Example: Timed automata

Timed automata were introduced by Rajeev Alur at Stanford during his PhD thesis under David Dill.



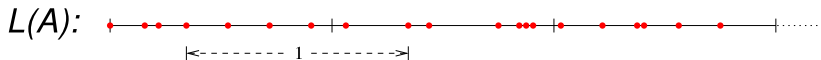
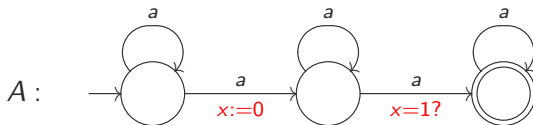
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Automata with clocks that run over $\mathbb{R}_{\geq 0}$, and clock constraints that determine which transitions are available.

Example: Timed automata



Adding time metrics to the models

We want to add a metric to the model so we can enforce certain timing constraints, for example:

*“Apply brake pads between 5 to 10 time units
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Note

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ all have suitable distance metrics.

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 MTL = LTL + **timing constraints on operators**:

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Formally,

$$\begin{aligned} \theta ::= & P \mid \theta_1 \wedge \theta_2 \mid \theta_1 \vee \theta_2 \mid \neg \theta \\ & \mid \mathbf{F}_I \theta \quad (\theta \text{ occurs in the Future in the interval } I) \\ & \mid \mathbf{G}_I \theta \quad (\theta \text{ occurs always (Globally) in the interval } I) \\ & \mid \theta_1 \mathbf{U}_I \theta_2 \quad (\theta_2 \text{ holds in } I \text{ and Until then } \theta_1 \text{ holds}) \end{aligned}$$

where I is an interval of $\mathbb{Z}$.

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We can also add timing constraints to past modalities.

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Formally

$$\begin{aligned} \pi &::= x + c \mid c \\ \theta &::= P \mid \pi_1 \leq \pi_2 \mid \pi_1 \equiv_d \pi_2 \mid x.\theta \\ &\quad \mid \neg\theta \mid \theta_1 \wedge \theta_2 \mid \theta_1 \vee \theta_2 \\ &\quad \mid \mathbf{F} \theta \mid \mathbf{G} \theta \mid \theta_1 \mathbf{U} \theta \\ &\quad \mid \mathbf{P} \theta \mid \mathbf{H} \theta \mid \theta_1 \mathbf{S} \theta \end{aligned}$$

Metric predicate logic

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*“Apply brake pads between 5 to 10 time units
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becomes

$$\forall x. \text{PEDAL}(x) \rightarrow \exists y. (x+5 < y < x+10) \wedge \text{BRAKE}(y),$$

a formula of $\text{FO}(<, +1)$.

Expressiveness and Decidability

Immediately we have

$$\text{LTL} \subseteq \text{MTL} \subseteq \text{MTL} + \text{Past} \subseteq \text{TPTL} \subseteq \text{FO}(<, +1) \subseteq \text{MSO}(<, +1)$$

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Over the integers $+1$ is definable in $\text{FO}(<)$:

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So $\text{LTL} = \text{FO}(<, +1)$ over $\mathbb{N}$ and $\mathbb{Z}$.

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So $\text{LTL} = \text{FO}(<, +1)$ over $\mathbb{N}$ and $\mathbb{Z}$.

Therefore:

Theorem

$\text{FO}(<, +1)$ satisfiability is decidable over $\mathbb{N}$ and $\mathbb{Z}$.

Decidability

Unfortunately:

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Corollary

FO($<, +1$) satisfiability is undecidable over $\mathbb{R}$

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Theorem (D'Souza, Holla & Vankadaru 2007)

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In $\text{MTL} + \text{Past}$:

$$\mathbf{G}_{=1}(\neg b \mathbf{S}_{(0,1)} a).$$

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Corollary

$\text{MTL} + \text{Past}$ is strictly more expressive than MTL.

Expressiveness in continuous time

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$$\varphi = \mathbf{F}_{\leq 1}(a \wedge \mathbf{F}_{\leq 1}a) \vee (\mathbf{F}_{\leq 1}a \wedge \mathbf{F}_{[1,2]}a) \vee \mathbf{G}_{=2}\mathbf{P}_{\leq 1}(a \wedge \mathbf{P}_{\leq 1}a)$$

Expressiveness in continuous time

$$\text{LTL} \subseteq \text{MTL} \subset \text{MTL} + \text{Past} \subset \text{TPTL} = \text{FO}(<, +1) \subseteq \text{MSO}(<, +1)$$

Theorem

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Question

What if we could “scale” the timing constants?

Temporal logics: $MTL_{\mathbb{Q}}$

$MTL_{\mathbb{Q}} = MTL + \text{Past} + \text{rational intervals}$

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Formally,

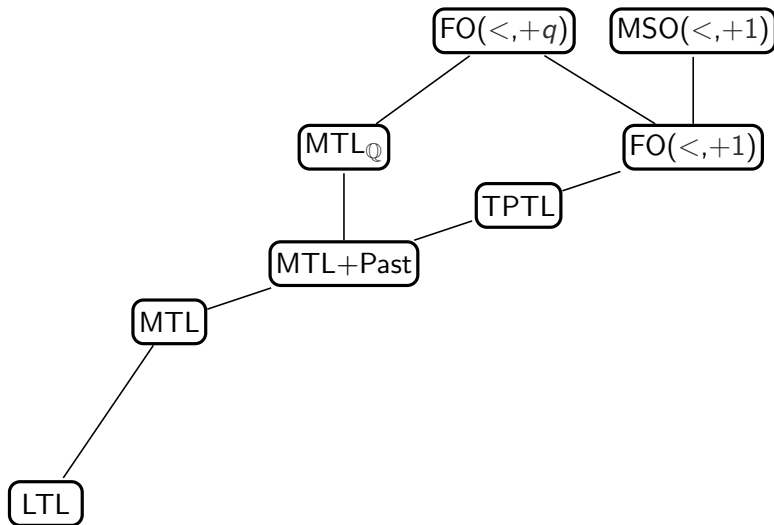
$$\begin{aligned} \theta ::= & P \mid \theta_1 \wedge \theta_2 \mid \theta_1 \vee \theta_2 \mid \neg \theta \\ & \mid \mathbf{F}_I \theta \quad (\theta \text{ occurs in the Future in the interval } I) \\ & \mid \mathbf{G}_I \theta \quad (\theta \text{ occurs always (Globally) in the interval } I) \\ & \mid \theta_1 \mathbf{U}_I \theta_2 \quad (\theta_2 \text{ holds in } I \text{ and Until then } \theta_1 \text{ holds}) \end{aligned}$$

where I is an interval of $\mathbb{Q}$.

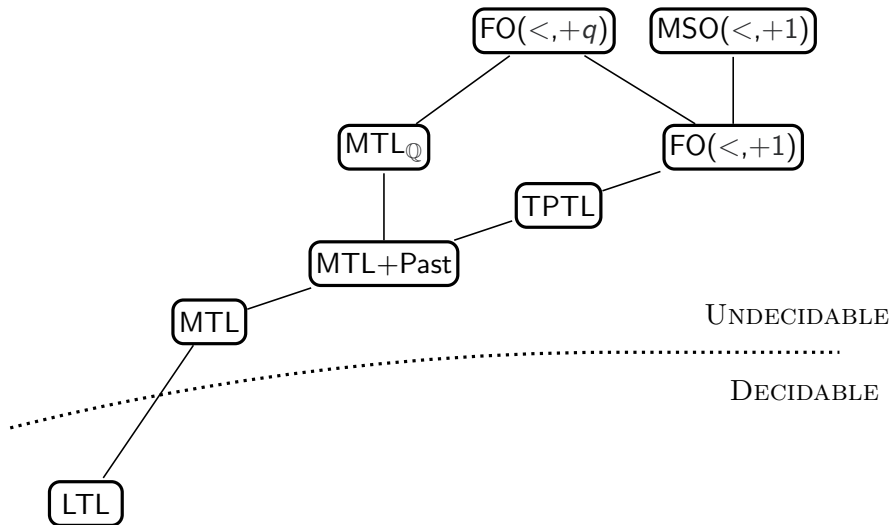
Predicate logics: $FO(<, +q)$

To allow for rational (or real) clock constants, we add to $FO(<)$ a *family* of unary functions: $+q$ for every $q \in \mathbb{Q}$ (or $\mathbb{R}$).

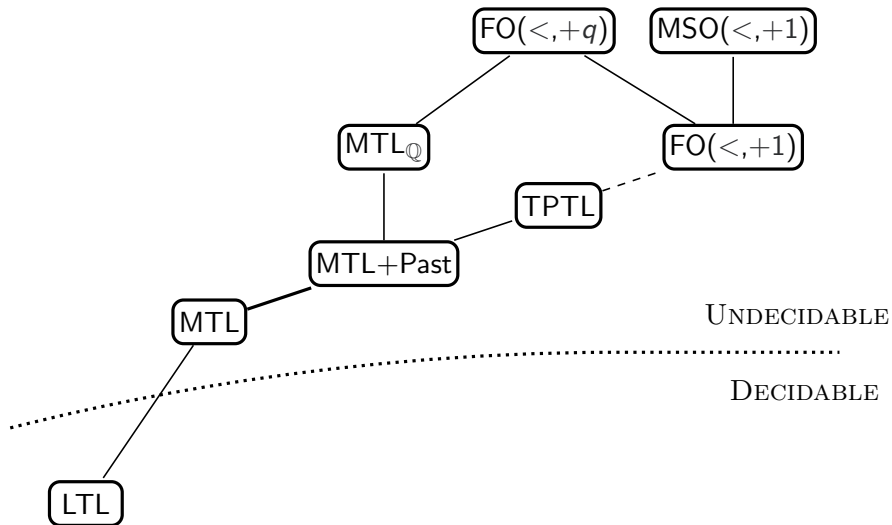
Temporal logic: Summary so far



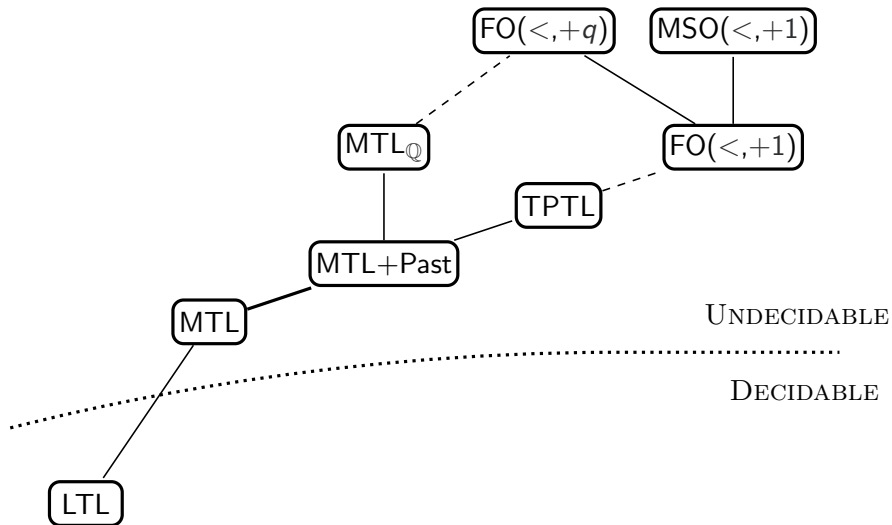
Temporal logic: Summary so far



Temporal logic: Summary so far



Temporal logic: To come...



Two results

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A metric temporal formula is:

- **pure distant past** if it is invariant on flows that agree on $(-1, -\infty)$
- **pure distant future** if it is invariant on flows that agree on $(1, \infty)$
- **bounded** if there is an N such that it is invariant on all flows that agree on $(-N, N)$

A temporal logic is *metrically separable* if every formula is equivalent to a boolean combination of pure distant past, pure distant future and bounded formulas.

A separation theorem for metric temporal logic

Theorem (H., Ouaknine, Worrell 2012)

Let $\mathcal{L}$ be a temporal logic.

- 1 If $\mathcal{L}$ is expressively complete then $\mathcal{L}$ is metrically separable.
- 2 If $\mathcal{L}$ is metrically separable then $\mathcal{L}$ is expressively complete if and only if it is expressively complete for all bounded formulas.

Expressive completeness of $MTL_{\mathbb{Q}}$

Theorem (H., Ouaknine, Worrell 2012)

$MTL_{\mathbb{Q}}$ is as expressive as $FO(<, +q)$.

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Theorem (H., Ouaknine, Worrell 2012)

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Proof sketch:

- Show it holds for bounded formulas.
- Use separation!

Weekly Feedback

I would appreciate any comments/suggestions/requests you have on this week's lectures (and tutorials).



<https://forms.office.com/r/0QhTKwtqda>